

1. Functional Dependency (FD: $X \rightarrow Y$)

Schema Invariant

Formal Rule: In relation R , $X \rightarrow Y$ holds iff $\forall$ tuples $t_1, t_2 \in R$:
If $t_1[X] = t_2[X] \Rightarrow t_1[Y] = t_2[Y]$ **MUST** equal $t_2[Y]$.

Determinant: X (LHS) $\xrightarrow{\text{[Decides]}}$ Dependent: Y (RHS)

- Instance vs Schema:** FD schema rule hota hai, data ke coincidence par depend nahi karta!
- Closure Property:** Har relation instance me integrity enforce karta hai.

2. FD Taxonomy & Classification

Core Theory

FD Type	Formal Condition	Example & Meaning
Trivial	$Y \subseteq X$	$\{\text{Roll, Name}\} \rightarrow \text{Roll}$ (Always true)
Non-Trivial	$X \cap Y = \emptyset$	$\text{Roll} \rightarrow \text{Name}$ (Real constraint)
Semi-Trivial	$Y \not\subseteq X \ \& \ X \cap Y \neq \emptyset$	$\{\text{Roll, Sub}\} \rightarrow \{\text{Sub, Marks}\}$
Full FD	$\forall X' \subset X, X' \not\rightarrow Y$	Y depends on FULL composite key (2NF)
Partial FD	$\exists X' \subset \text{Key}, X' \rightarrow Y$	Non-prime depends on part of key (Violates 2NF!)

3. Armstrong's Axioms (Sound & Complete Axiomatic Inference System)

F⁺ Generation

Primary Axiom 1: Reflexivity

If $Y \subseteq X \Rightarrow X \rightarrow Y$

Subset property: Har attribute set apne sub-parts ko trivially determine karta hai.

Primary Axiom 2: Augmentation

If $X \rightarrow Y \Rightarrow XZ \rightarrow YZ \ (\forall Z)$

Additive property: Dono taraf koi bhi attribute jod sakte hain.

Primary Axiom 3: Transitivity

If $X \rightarrow Y \ \& \ Y \rightarrow Z \Rightarrow X \rightarrow Z$

Chaining property: Dependency aage chain banakar transfer hoti hai.

Derived 4: Union / Additive

$X \rightarrow Y \ \& \ X \rightarrow Z \Rightarrow X \rightarrow YZ$

LHS same ho toh RHS combine ho sakta hai.

Derived 5: Decomposition

$X \rightarrow YZ \Rightarrow X \rightarrow Y \ \& \ X \rightarrow Z$

RHS split hota hai. LHS KABHI split nahi hota!

Derived 6: Pseudo-Transitivity

$X \rightarrow Y \ \& \ WY \rightarrow Z \Rightarrow WX \rightarrow Z$

Useful for proving implied dependencies.

Soundness: Derived FD hamesha physically valid hogi. | **Completeness:** Saari logically implied FDs (Closure F^+) nikal aayengi.

4. Attribute Closure Algorithm (X^+)Polynomial $O(|F| \cdot |R|)$

- Init:** Set $\text{closure} = X$.
- Loop:** Repeat for each $Y \rightarrow Z \in F$:
If $Y \subseteq \text{closure}$, then $\text{closure} = \text{closure} \cup Z$.
- Fixed-Point:** Loop tab tak chalao jab tak koi naya attribute add na ho.
Return closure as X^+ .

Key Applications:

- Check if $A \rightarrow B$ holds: Test if $B \subseteq A^+$.
- Super Key Test: Check if X^+ contains ALL attributes ($X^+ = R$).

5. Fast Candidate Key Discovery (3-Category Matrix)

AKTU Exam Cheat

Category	Attribute Position	Key Presence Rule
Category A (Essential)	NEVER on RHS	MUST be in EVERY Candidate Key!
Category B (Pivot)	Both on LHS & RHS	Test combinations with Category A.
Category C (Extraneous)	ONLY on RHS (Never LHS)	CANNOT be in ANY Candidate Key!

Prime Attribute: Member of ANY Candidate Key.

Non-Prime Attribute: NOT part of ANY Candidate Key.

6. Equivalence of FD Sets ($F \equiv G$)

F* = G* Test

Do sets F aur G tabhi equivalent hote hain jab dono ek doosre ko cover karein:

- Test 1 (G covers F):** $\forall X \rightarrow Y \in F$, check if $Y \subseteq X^+_G$.
- Test 2 (F covers G):** $\forall A \rightarrow B \in G$, check if $B \subseteq A^+_F$.

Both tests pass $\Rightarrow F \equiv G$. Polynomial time verification!

8. The 3 Database Anomalies

Root: Redundancy

- Insertion Anomaly:** Cannot insert an independent entity without a primary key (NULL not allowed).
- Deletion Anomaly:** Deleting one tuple inadvertently destroys unrelated secondary entity data.
- Update Anomaly:** Redundant copies cause data inconsistency if some rows fail to update.

7. Canonical Minimal Cover (Fc) 3-Step Method

Irreducible Set

- Singleton RHS:** Break $X \rightarrow ABC$ into $X \rightarrow A$, $X \rightarrow B$, $X \rightarrow C$.
- Remove Extraneous LHS:** In $AB \rightarrow C$, if $C \in (A)^+$ without B, drop B!
- Remove Redundant FDs:** For each $X \rightarrow Y$, test if $Y \in X^+_{(F - \{X \rightarrow Y\})}$. If YES, delete!

9. First Normal Form (1NF)

Atomic Values

- Rule:** All attribute domains must contain strictly **atomic (indivisible) values**.
- Eliminate Multi-valued attributes (lists/arrays in cells).
 - Decompose Composite attributes ($Address \rightarrow City, PIN$).
 - Eliminate Repeating groups (numbered columns like Skill_1, Skill_2).

10. Core Normal Forms In-Depth (2NF, 3NF, BCNF)

Master Decision Matrix

Normal Form	Prerequisite	Formal Condition to Satisfy	Violation Signature	How to Decompose
2NF	1NF	No Partial Dependency: Every non-prime attribute must depend on the FULL candidate key.	Proper Subset of CK $\rightarrow$ Non-Prime	$R1(X', Y)$ with key X' , and $R2(R - Y)$.
3NF	2NF	No Transitive Dependency: For every non-trivial $X \rightarrow Y$: X is Super Key OR Y is Prime Attribute	Non-Key $\rightarrow$ Non-Prime (LHS not SK and RHS not Prime)	Bernstein Synthesis: $R_i = X \cup Y$ from Canonical Cover Fc. Add CK if needed.
BCNF	1NF	Strict Super Key: For every non-trivial $X \rightarrow Y$: X MUST BE A SUPER KEY OF R (Zero Prime exceptions!)	LHS is NOT a Super Key (Even if RHS is Prime!)	$R1(X \cup Y)$ and $R2(R - Y)$. Guaranteed lossless, but may lose FD preservation!

11. Special Theorem: Simple Candidate Keys

Speed Shortcut

Theorem: Agar kisi relation ke sabhi candidate keys **Single Attribute (Simple)** hain, toh wo relation **AUTOMATICALLY 2NF** me hota hai!

Proof: Single attribute ka koi non-empty proper subset nahi hota, isliye partial dependency mathematically impossible hai!

12. The Great 3NF vs BCNF Trade-Off

Industry Standard

- 3NF:** Always guarantees **Lossless Join AND Dependency Preservation**. Slight redundancy allowed.
- BCNF:** Eliminates ALL functional redundancy, but **MAY LOSE functional dependencies!**

13. Lossless Join (Non-Additive) Decomposition

Mandatory $R = R_1 \bowtie R_2$

Binary Decomposition Theorem: Split of R into R_1, R_2 is Lossless iff:

$(R_1 \cap R_2) \rightarrow R_1 \text{ OR } (R_1 \cap R_2) \rightarrow R_2$

Intuition: Common attribute set MUST be a **Super Key in at least one** table!

If this condition fails $\Rightarrow$ **Spurious Tuples** (fake records) appear upon natural join!

15. Multivalued Dependencies (MVD) & 4NF

Independent Multi-Facts

MVD Definition ($X \twoheadrightarrow Y$): Y values associated with X are independent of Z values. Causes $m \times n$ Cartesian explosion!

$t1[X]=t2[X] \Rightarrow \exists t3, t4$ with swapped Y and Z coordinates

4NF Condition: Relation in BCNF, and $\forall$ non-trivial $X \twoheadrightarrow Y$: **X MUST BE A SUPER KEY OF R.**

Fagin's Split: $R(X,Y,Z)$ decomposes into $R1(X,Y)$ and $R2(X,Z)$.

14. Dependency Preservation Property

$(F1 \cup \dots \cup Fk)^+ = F^+$

Decomposition is dependency preserving if all original constraints can be checked within individual tables without executing expensive cross-table joins.

- $Fi = \pi_{Ri}(F) = \{ X \rightarrow Y \in F^+ \mid XY \subseteq Ri \}$
- 3NF:** Always preserves all FDs ($\checkmark$).
- BCNF:** May not preserve some FDs ($\times$).

16. Join Dependencies (JD) & 5NF (PJNF)

Ultimate Projection Form

Join Dependency: $\bowtie[R1, R2, \dots Rn]$ holds if $R = \bowtie \pi_{Ri}(R)$. Models 3-way cyclic relationships.

5NF Condition: In 4NF, and for every non-trivial JD: **EVERY Ri MUST BE A SUPER KEY OF R.**

Inclusion Dependency (IND): $R[X] \subseteq S[Y]$.
Theoretical foundation of **Foreign Key & Referential Integrity!**

17. Universal AKTU 10-Mark Normalization Solving Algorithm

Guaranteed Full Marks

Step 1: Find All Candidate Keys

- Partition attributes: Cat A (Never RHS), Cat B (Both), Cat C (Never LHS).
- Compute $(Cat A)^+$. If equal to R $\Rightarrow$ Unique Candidate Key!
- If not, combine with Cat B singles/pairs. Identify Prime vs Non-Prime.

Step 2: Test 2NF (Partial Dependency)

- Is any Non-Prime attribute determined by a proper subset of CK?
- If YES $\Rightarrow$ Violates 2NF (Highest NF = 1NF). Decompose into $R1(X, Y)$ & $R2(R - Y)$.

Step 3: Test 3NF (Transitive Dependency)

- For every $X \rightarrow Y$: Is X a Super Key? OR Is Y a Prime Attribute?
- If neither holds $\Rightarrow$ Violates 3NF. Highest NF = 2NF. Decompose via Bernstein.

Step 4: Test BCNF (Strict Super Key)

- For every $X \rightarrow Y$: Is X strictly a Super Key?
- If NO $\Rightarrow$ Highest NF = 3NF. Decompose R into $R1(X \cup Y)$ & $R2(R - Y)$.

18. Master Normal Form Inclusion & Anomaly Elimination Summary

Complete Hierarchy

Normal Form	Hierarchy Inclusion	Specific Anomaly Eliminated	Exam Test Rule
1NF	Base Relational	Non-atomic domains, multi-valued cells	All cell values atomic
2NF	$2NF \subset 1NF$	Partial dependency on composite keys	No Proper Subset of CK $\rightarrow$ Non-Prime
3NF	$3NF \subset 2NF$	Transitive dependency through non-keys	LHS Super Key OR RHS Prime Attribute
BCNF	$BCNF \subset 3NF$	Anomalies from overlapping candidate keys	LHS MUST be Super Key
4NF	$4NF \subset BCNF$	Cartesian explosion of independent facts	All MVD determinants are Super Keys
5NF (PJNF)	$5NF \subset 4NF$	N-way join spurious tuples & redundancy	All JD components are Super Keys